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ADVANCED MICROECONOMICS

Behavioral Economics - a teaser

Winter Term 2025/2026

Outline

1) **Consumer Theory**
   1.1) Preferences and Utility (L1)
   1.2) Utility Maximization (L2)
   1.3) Expenditure Minimization (L3)
   1.4) Demand Functions and Duality (L4+L5)
2) **General Equilibrium**
   2.1) Exchange Economy (L6)
   2.2) Welfare Economics and General Equilibrium (L7)
3) **Decisions under Risk and Uncertainty**
   3.1) Expected Utility (L8)
   3.2) Risk Preferences (L9)
   3.3) Insurances (L10)
   3.4) Implicit Markets for Risk and the Value of a Statistical Life (11)
4) **Behavioral Economics** (L12)

Overview

Literature

- Angner, Erik (2021): A Course in Behavioral Economics, 3rd edition, Palgrave McMillan, Chapter 3 and 7
- Thaler, R. H. (2016). Behavioral economics: Past, present, and future. American Economic Review, 106(7), 1577-1600.
- Kahneman, D., & Tversky, A. (1979). Prospect theory: An analysis of decision under risk. Econometrica, 47(2), 263–292.
- Kahneman, D. (2002). Maps of bounded rationality: A perspective on intuitive judgement and choice. Noble Prize Lecture. December 8, 2002
- Kahneman, D., Knetsch, J. L., & Thaler, R. H. (1990). Experimental tests of the endowment effect and the Coase theorem. Journal of Political Economy, 98(6), 1325–1348.

Overview

What is the problem of the standard theory?

- Recap: The goal of neoclassical consumer theory (under certainty) and expected utility theory (under risk) is the **description** of the behavior of rational consumers
- One theory to reach two goals:
  1. Characterize what people should do optimally **(normative theory)**
  2. Predict what people actually do **(positive/descriptive theory)**
- While standard axiomatic theory is valid for the first goal, it fails for the second

Why is a standard theory not a positive theory?

- Standard theory starts with axioms/assumptions that people in real life do not fulfill
- People
  - have well-defined preferences and unbiased beliefs
  - make optimal choices based on these preferences and beliefs (and thus have infinite cognitive abilities)
  - care about themselves
- **But:** In reality, the underlying assumptions are often violated: people ...
  - ... violate transitivity
  - ... do not consider all possible alternatives and their opportunity costs
  - ... overweight certain (safe) alternatives
  - ... react more strongly to losses than to gains

→ The descriptive validity of rational choice theory is limited *(even in situations without uncertainty)*

What is behavioral economics?

- Assumptions of rational choice theory describe an **"Homo Oeconomicus"** (or an Econ), but we are **"Humans"** with biased beliefs, limited cognitive abilities that do not always do what is best for us
- **Standard theory studies econs in abstract situations, not humans in real-life**

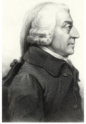
Behavioral Economics

Identification of systematic deviations from the rational choice model by integrating psychology into economic analysis to better understand and model human behavior.

Overview

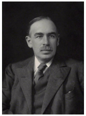
History of behavioral economics

- **Behavioral economics is not new!**
- **Adam Smith** described concepts that we'd today call overconfidence, loss-aversion, and self-control problems
- **Arthur Pigou** described present-bias (time inconsistency in intertemporal choices)
- **John Maynard Keynes** started *"behavioral finance"*
    - Descriptive part got lost after WW II, when economics became more rigorously math-based



Adam Smith
(1723 - 1790)

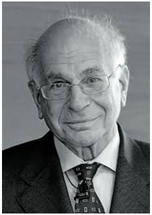
Arthur Pigou
(1877 - 1959)



John M. Keynes
(1883 - 1946)

Heuristics and Biases Program

- The foundations of modern behavioral economics were laid in the 1970s, mainly through the work of psychologists Daniel Kahneman and Amos Tversky
- **Heuristics and Biases** – research program developed by Kahneman and Tversky
- Identification of deviations/anomalies in actual behavior relative to the predictions of the neoclassical model
- Can be broadly grouped into two categories:
  1. **Heuristics**: mental shortcuts or experience-based rules of thumb used under time pressure or incomplete information
  2. **Cognitive biases**: systematic errors in thinking



Daniel Kahneman
(1934 - 2014)
Nobel Laureate 2002

Amos N. Tversky
(1937 - 1996)

Behavioral Economics after Kahneman & Tversky

- From the 1980s onward: Integration of psychological insights into **mainstream economics**

→ Behavioral economics becomes part of the **economic core**, not a fringe approach

- Central figure: **Richard Thaler**
  - Focus on **systematic behavioral patterns** in economic decision-making
  - Key contributions: Mental accounting, Self-control problems and present bias, Limited attention and bounded rationality
  - **Behavioral Policy** Strong emphasis on policy relevance

  → **Libertarian paternalism and Nudging**
    - "Nudges" can be used to push human decisions into specific directions

Richard Thaler
(1945 -)
Nobel Laureate 2017

Overview

1. Standard Theory vs. Behavioral Economics
   - Limits of the standard theory
   - Behavioral Economics
2. History of Behavioral Economics
3. **Contributions of Behavioral Economics**
   - Explain-away-tions
   - Experimental Economics
4. Central Concepts of Behavioral Economics
   - Heuristics and Biases
   - Probability Judgment
   - Prospect Theory
5. Conclusion and Outlook to Summer Term

Overview

Explanations or Explain-away-tions?

- Richard Thaler in an interview with Katherine Milkman (available here) on the question whether he has any thoughts on how to **convince traditional economists** to change their views:

> *"The truth is that I don't think I've changed anybody's mind [...] In some ways, I succeeded by not trying to change everybody's minds, but by recruiting others [young people] to help build up a* **body of evidence***. [...] The only thing that convinces anybody is data. So there were lots of excuses economists made for continuing with their ways. Matthew Rabin sometimes refers to these as* **"explain-away-tions"***."*

"Explain-away-tions"

- "Fight" between behavioral economists/psychologists (like Thaler or Tversky/Kahneman) and theorists
- Arguments that were put forward by the standard theory proponents
  - Standard theory is only meant **"as if"** (we do not think people actually do the maximization, the outcomes are just as if they did it)
  - Results of behavioral economists built on **hypothetical questions and small stakes**, and would further go away if people were faced with these questions more often
  - Biases would be corrected through **markets**

"Evidence-based Economics"

- Thaler's data-driven approach follows the trend to turn economics into an **"evidence-based discipline"**
- Behavioral economists document systematic deviations in human decisions compared to predictions by standard theory and then help to adapt or replace standard theory to develop a better descriptive theory
- Examples
  - Prospect Theory (Kahneman and Tversky, 1979) adapts Expected Utility Theory to
    - allow for reference-dependence and loss aversion
    - biased beliefs about probabilities
  - Social preferences
    - utility functions to not only depend on own payoff/consumption but also on that of other people

Overview

Experimental Economics

- Experimental economics became a central approach of **evidence-based economics**:
  - provides **clean tests** of standard vs. behavioral models ("data beats explain-away-tions")
  - allows to **identify causal effects** by design (not by assumptions)
  - helps uncover **mechanisms**: beliefs, preferences, social motives, mistakes

Experimental Economics

Experiments

Controlled studies in which researchers manipulate a variable (treatment) and compare outcomes to a control group.

- **Key idea:** isolate the effect of a treatment by holding other factors constant
- **Randomization:** makes treatment and control comparable
  $\Rightarrow$ differences in outcomes have a **causal interpretation**
- **Two main settings:**
  - **Laboratory experiments:** high control, clean measurement, stylized environments
  - **Field experiments / RCTs:** real-world context, higher external validity

Overview

Overview: Central concepts (very short)

Behavioral economics documents systematic deviations from standard theory and builds models that predict behavior better.

1. **Decisions under certainty:** limited attention, context dependence, reference points
2. **Decisions under risk:** biased probability judgments, framing, prospect theory
3. **Intertemporal decisions:** present bias, self-control, prediction errors *[not today]*

Goal of this teaser: give you a map of the "toolbox" we will use later.

Overview

Bounded rationality and limited attention

- **Rational Choice:** Stable preferences + consistent choice.
- **Reality:** decisions are often made under limited attention and cognitive constraints.
    - **Opportunity costs** (utility from best alternative) are neglected
- **Behavioral economics:** patterns are predictable (not random noise) and depend on context.
- **Implication:** we should study choice architecture (which options are presented how).

Sunk cost fallacy

Sunk Costs

Costs that have already been incurred and **cannot be recovered**, i.e. which are irreversible.

- **Rational Choice:** Irreversible past costs should not affect current choices.
- **Behavioral Economics:** people systematically consider irreversible costs to avoid feeling that past spending was wasted $\rightarrow$ Sunk cost fallacy / escalating commitment
- **Examples:** memberships and subscriptions, investment decisions

Menu dependence & the decoy effect

- **Rational Choice:** Adding an irrelevant alternative should not change choices (IIA).
- **Reality:** Adding an asymmetrically dominated option can shift demand to a "target".
- Intuition: people compare locally and use relative evaluation ("which looks like the best deal?").
- *Examples:* pricing of "basic / premium / premium+" plans, tip options

Framing effects

- **Framing:** Preferences/choices change when the same options are described differently
- Classic examples:
  - "half full" vs "half empty"
  - "save 80/100 jobs" vs "fire 20/100 workers"
- Violates the idea that only outcomes matter (description should not matter)

Reference points and Endowment effect

- **Reference-point phenomenon —** preferences may depend on **reference points** (e.g. endowment, status quo)
- Example: Preferences depend on what you own (your endowment / status quo).

Endowment Effect

The preferences over alternatives depend on what an individual owns in the initial situation (their "endowment").

- Empirical regularity: **willingness to accept** (WTA) $>$ **willingness to pay** (WTP).
- Related: **Anchoring effect** - People start decision process from an anchor (often irrelevant) and adjust insufficiently

Endowment Effect

Application — Kahneman, Knetsch, Thaler (1990)

- Laboratory experiment: Every second participant is randomly given a coffee mug (worth $6)
- Participants can then sell the mug to those who did not receive one
- Buyers state their maximum willingness to pay (WTP), sellers their minimum willingness to accept (WTA)
- The median buyer was willing to pay $2.75, while the median seller demanded $5.25 for the mug
- Being endowed with a coffee mug affects how valuable it is perceived by the individual

Source: Kahneman, D., Knetsch, J. L., & Thaler, R. H. (1990). Experimental tests of the endowment effect and the Coase theorem. *Journal of Political Economy*, 98(6), 1325–1348.

Loss aversion

- A reason for endowment effect could be aversion against losses
- **Core idea**: Outcomes are evaluated relative to a reference point (endowment, expectations, goals) and utility is not evaluated absolutely but **relatively**

Loss aversion

People suffer more from losing something than they benefit from gaining the same amount; that is, losses carry more weight than gains.

- **Consequence:** "Kink" of the utility function at the reference point (behavior changes around the status quo)

Overview

Risk: Subjective probabilities are biased

- **Expected Utility Theory** needs: (i) stable utility, (ii) correct probability beliefs.
- **Evidence:** probability judgments rely on heuristics $\Rightarrow$ systematic errors.
- Three (common) deviations from this assumption:
  - representativeness
  - availability
  - confirmation bias

Representativeness: base-rate neglect & gambler's fallacy

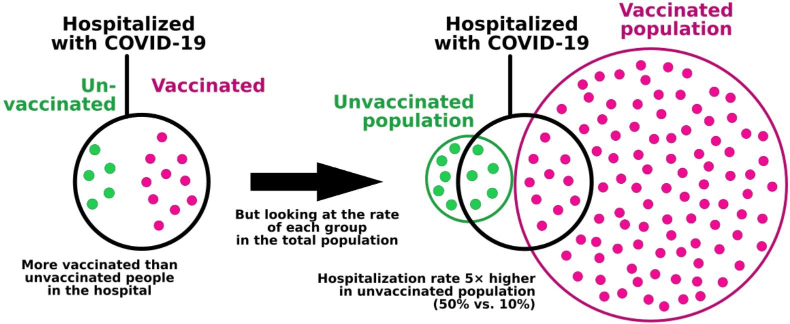
Representativeness Heuristic

People assess the probability of an event based on how *representative* or typical it appears for a known category, rather than according to statistical principles.

- "How typical does it look?" replaces statistical reasoning.
- Biases:
  - **Base-rate neglect:** ignoring prior probabilities
  - **Gambler's fallacy:** believing random sequences must "balance out" quickly
- **Implication:** misperception of risks (medical tests, lotteries, rare events).

Example for base-rate neglect

- **Example:** COVID-19 Hospitalizations



Source: Wikipedia

Availability & Confirmation bias

Availability Bias

People assess probabilities based on how easily examples come to mind, rather than on how frequently events actually occur.

Confirmation Bias

Confirmation bias describes the tendency of individuals to select, interpret, and recall information in a way that confirms their existing beliefs.

- **Implication:** risk perceptions can diverge strongly from objective probabilities.

Framing under risk: Asian Disease problem

- Application of **Framing effect** to probability judgments in loss/gain context
- **Asian Disease problem** by Kahneman & Tversky (1984) - Same expected outcomes of risky decision but different description:
  - **Gain frame:** "200/600 people will be saved" ⇒ many choose the sure option.
  - **Loss frame:** "400/600 people will die" ⇒ many switch to the risky option.
- **Takeaway:** the reference point (gains vs losses) changes risk taking
- Humans tend to be risk-averse in the gains and risk-loving in the losses frame

Certainty effect & the Allais paradox

- People overweight certainty: a sure gain gets extra value beyond its probability mass.
- This can violate the "sure-thing principle" (**Allais paradox**).

Allais Paradox

An irrelevant state of the world (a sure thing) influences choices between two options.

- **Intuition:** avoiding the regret of "ending up with nothing".
- **Implication:** Demand for insurance and lottery tickets can coexist (for risk-averse individuals).

Overview

Prospect theory

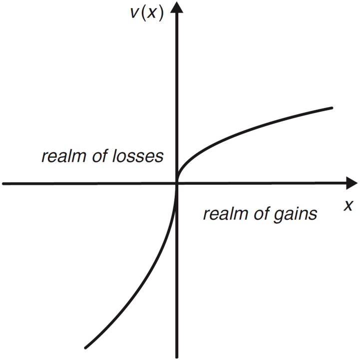
Prospect theory

Prospect theory is the core theory of behavioral economics (based on work by Kahneman & Tversky) and describes how individuals make decisions under risk and uncertainty, emphasizing that people tend to prioritize avoiding losses over acquiring equivalent gains.

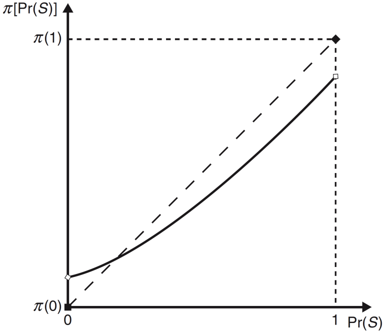
- Prospect theory modifies expected utility theory along two dimensions:
  - **Value function** $v(x)$ over gains/losses relative to a reference point
  - **Probability weights** $\pi(p)$ instead of objective probabilities p
- Explains: framing effects, loss aversion, certainty effect...

Prospect Theory

Value Function



Probability Weighting



Source: Angner, 2021, Figure. 7.2 and 7.6.

Risk Preferences in Prospect Theory

- Individuals' behavior under risk depends on whether
  1. ... outcomes are perceived as **gains or losses** and
  2. ... the relevant **probabilities are high or low**.

Objective Probability	Domain: Losses	Domain: Gains
Low	**risk-averse**	**risk-loving**
Medium or High	risk-loving	risk-averse

Adapted from Angner (2021), Table 7.5, p. 170

→ In the **domain of losses**, individuals tend to be risk-loving — except for lotteries involving a rare event with a strongly negative outcome (e.g. insurance) while in the **domain of gains**, individuals tend to be risk-averse — except for lotteries involving a rare event with a strongly positive outcome (e.g. gambling).

Overview

Take-away: What behavioral economics adds to microeconomics

- Behavioral economics keeps the micro toolkit (preferences, optimization), but changes:
  - **what enters utility** (reference points, social preferences, self-control)
  - **how beliefs are formed** (heuristics, probability weighting)
  - **how choices are constructed** (framing, menus, attention)

Standard Theory and Behavioral Economics

Thaler, R. H. (2016). Behavioral economics: Past, present, and future. American Economic Review, 106(7), 1577-1600.

A second general point is that we should not expect some new grand behavioral theory to emerge to replace the neoclassical paradigm. We already have a grand theory and it does a really good job of characterizing how optimal choices and equilibrium concepts work. Behavioral theories will be more like engineering, a set of practical enhancements that lead to better predictions about behavior. So far, most of these behavioral enhancements focus on two broad topics: preferences and beliefs.

→ Behavioral economics tries to describe how humans behave and aims at increasing the ability to predict human behavior which will e.g. help in design of policy instruments

Outlook: Behavioral and Experimental Economics (WIW.06732.03) in Summer 2026

- **Lecturer**: Dr. Juliane Hennecke (exercise sessions: TBA)
- **Goal:** Course addresses the question how well the assumptions about rational agents and well-defined preferences describe *actual* human behavior by...
  - studying **systematic deviations** from rational choice
  - integrating insights from **psychology** into economic models
  - testing theories using **laboratory and field experiments**

  Content:
  - Heuristics and biases, prospect theory, present bias
  - Social preferences, reciprocity, and trust
  - How experiments are designed, analyzed, and interpreted
  - Applications to public policy and firms

If you enjoyed Advanced Micro, this course shows you where theory meets real behavior.

Keywords

- Heuristics
- Cognitive Biases
- Loss aversion
- Reference points
- Endowment effect
- Framing
- Prospect theory

Questions?